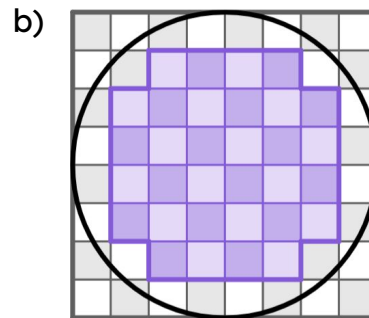
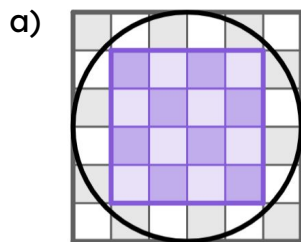




Area of a circle

Task A: Estimating areas of circles

- 1) For each diagram: i) find the number of complete squares inside the circle;
 ii) find the total number of squares;
 iii) complete the range for the area of the circle.



i) number of squares inside =

i) number of squares inside =

ii) total number of squares =

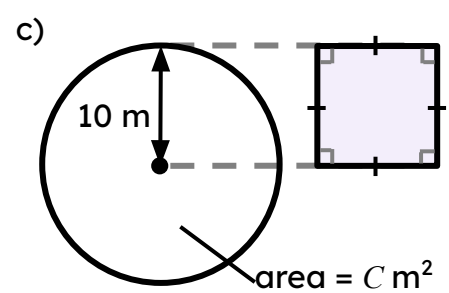
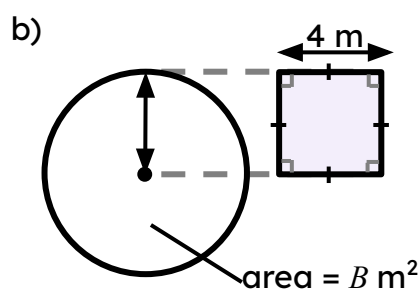
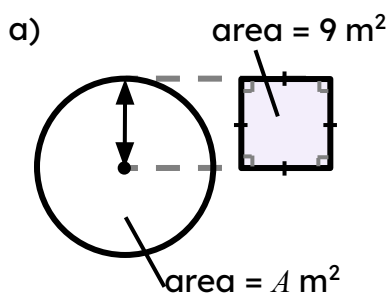
ii) total number of squares =

iii) ___ units² < area of circle < ___ units²

iii) ___ units² < area of circle < ___ units²

- 2) Use the information provided to fill in the gaps.

The area of each circle is between 2 and 4 times the area of its accompanying square.



area of 2 squares = ___ m²

area of 2 squares = ___ m²

area of 2 squares = ___ m²

area of 4 squares = ___ m²

area of 4 squares = ___ m²

area of 4 squares = ___ m²

___ < A < ___

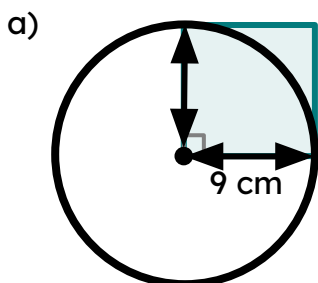
___ < B < ___

___ < C < ___

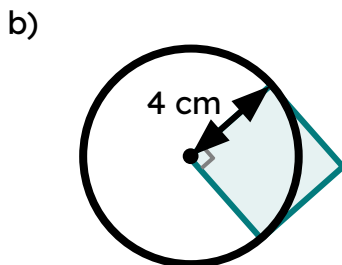
Task B: Deriving the formula to calculate areas of circles

1) Each circle overlaps with a square. For each question:

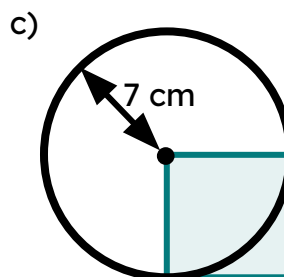
- Find the area of the square.
- Find the area of the circle in terms of π .



- area of square =
- area of circle =

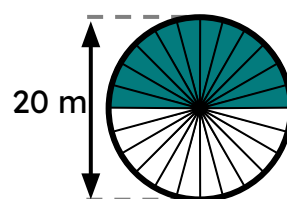
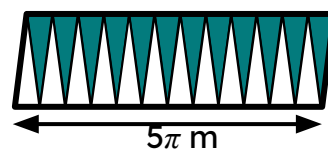
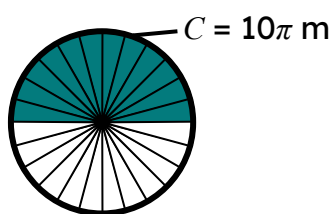
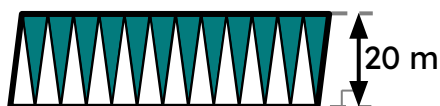
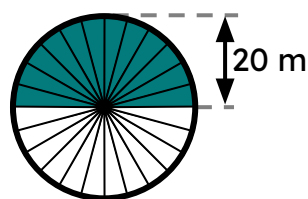
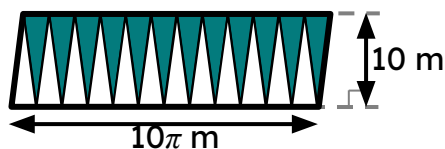


- area of square =
- area of circle =



- area of square =
- area of circle =

2) Each parallelogram is made from the sectors of one of the three circles. The diagrams are not drawn to scale.



- Match each parallelogram with the circle of the same approximate area.
- Find the area of each circle in terms of π .

